Building Semantic Search using Elastic & Jina

Overview of Elastic Search , Kibana Dashboard, Building Agents using Jina

By:

Crystal Darling B

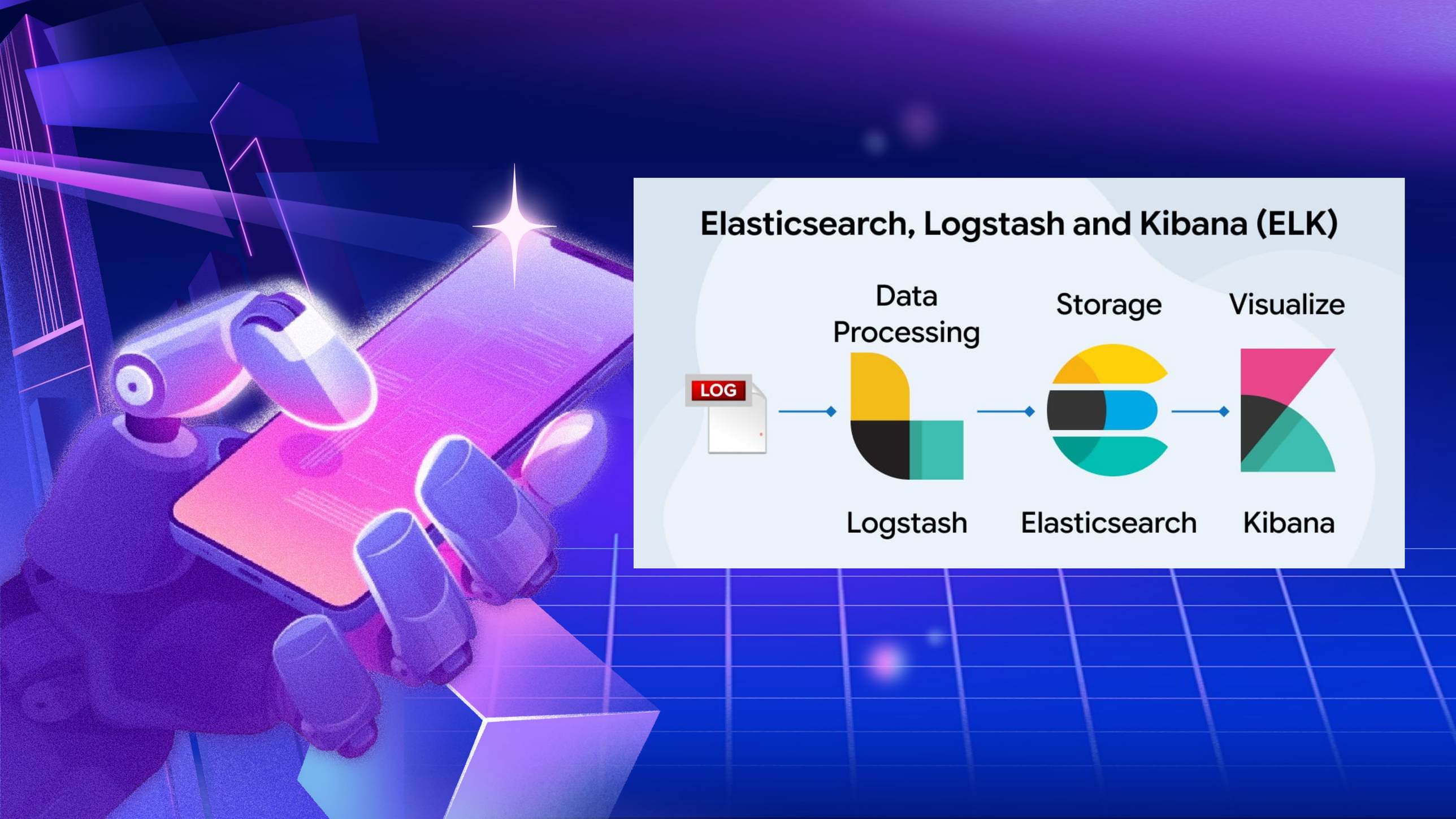
Lead & Organizer of AI Geeks Chennai

What is Elastic Search?

The Ultimate Search & Analytics Engine





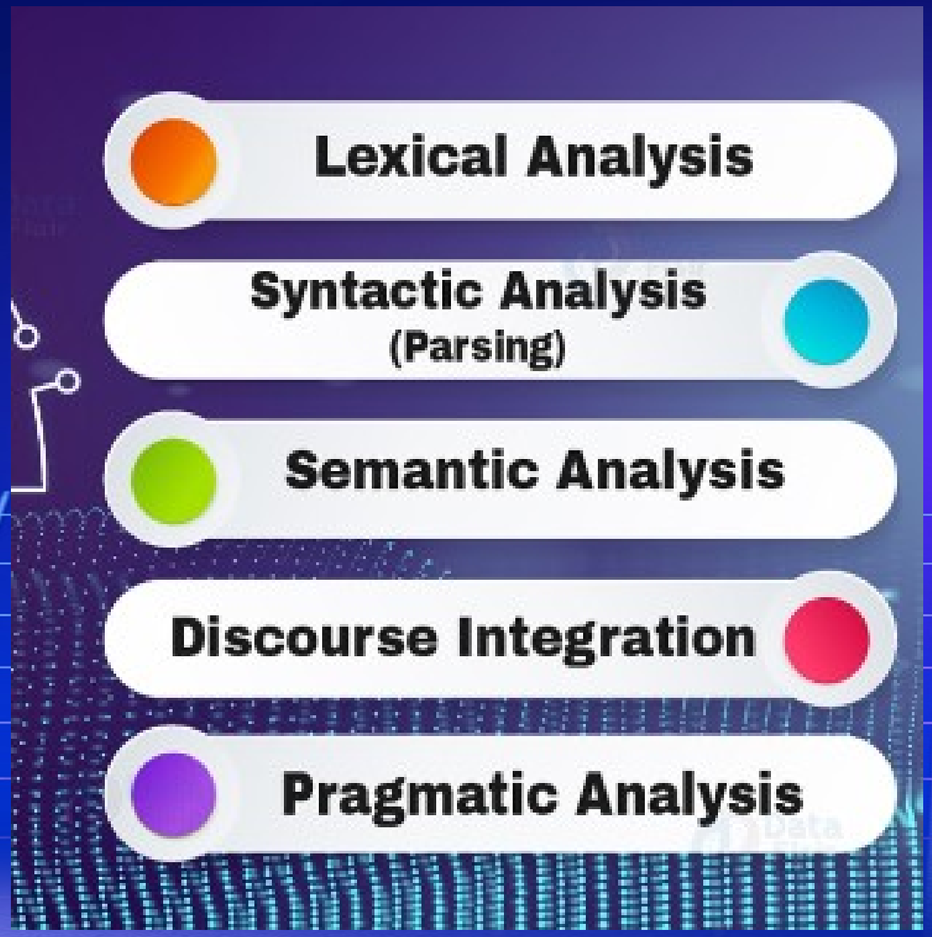

Elasticsearch, Logstash and Kibana (ELK)
Data Processing
Storage
Visualize
LOG
Logstash
Elasticsearch
Kibana

What is Jina?



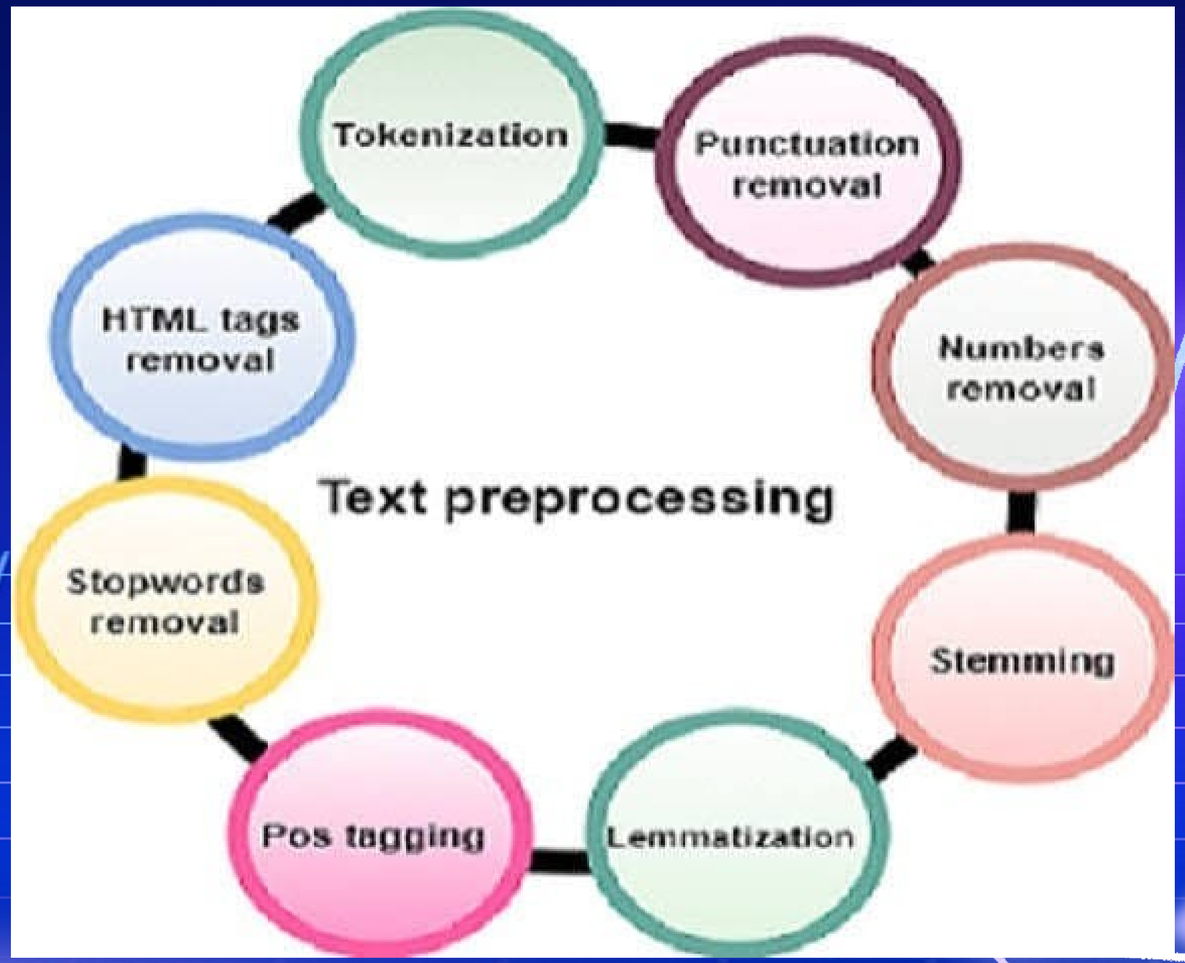


Steps in Natural Language Processing





Tokenization / Stemming / Lemmatization







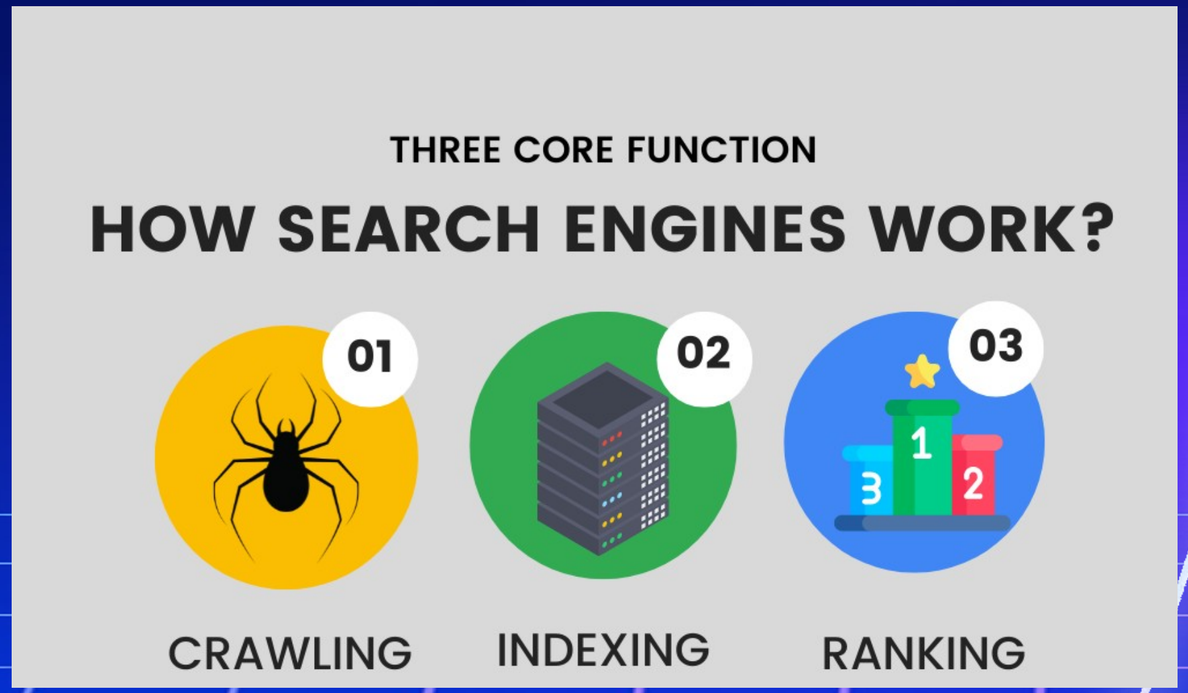
SEMANTIC VS LEXICAL SEARCH



SHAPING USER EXPERIENCE ONLINE

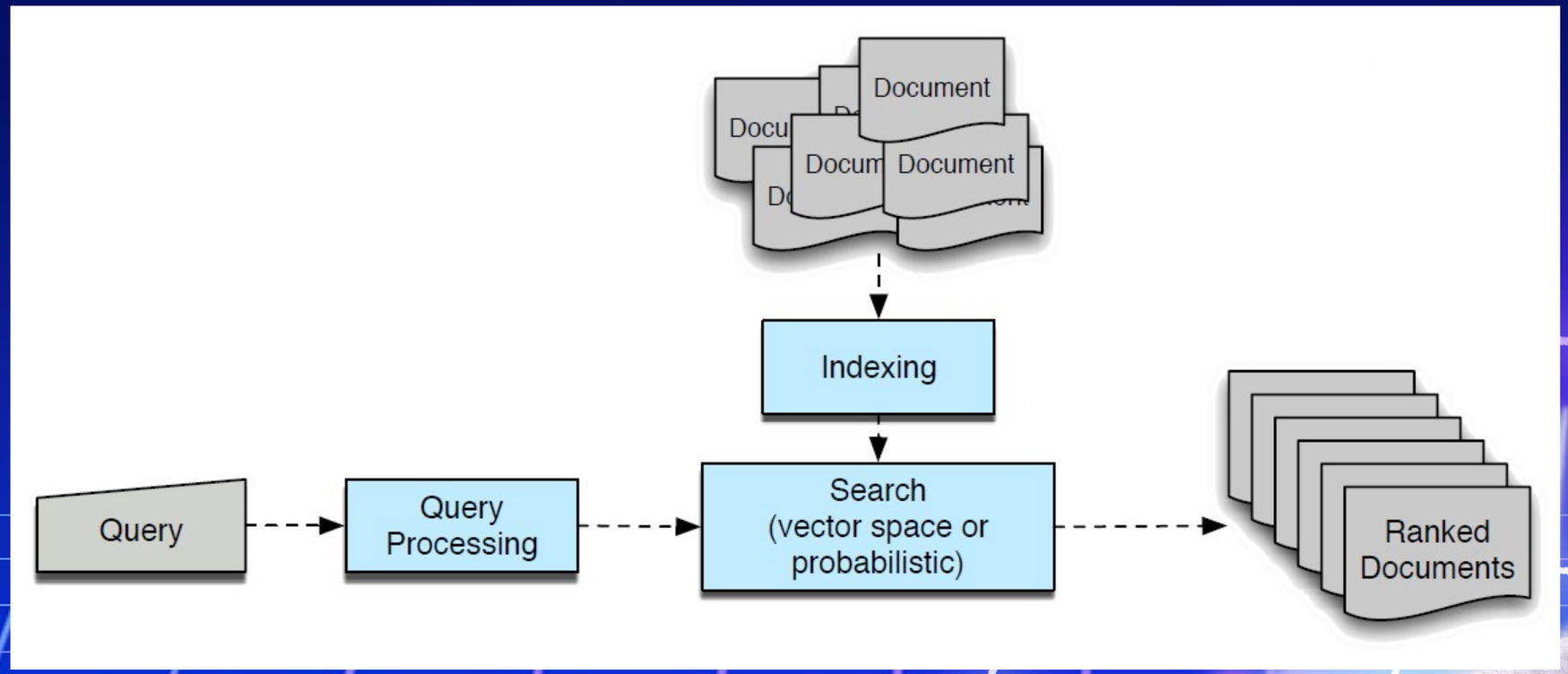


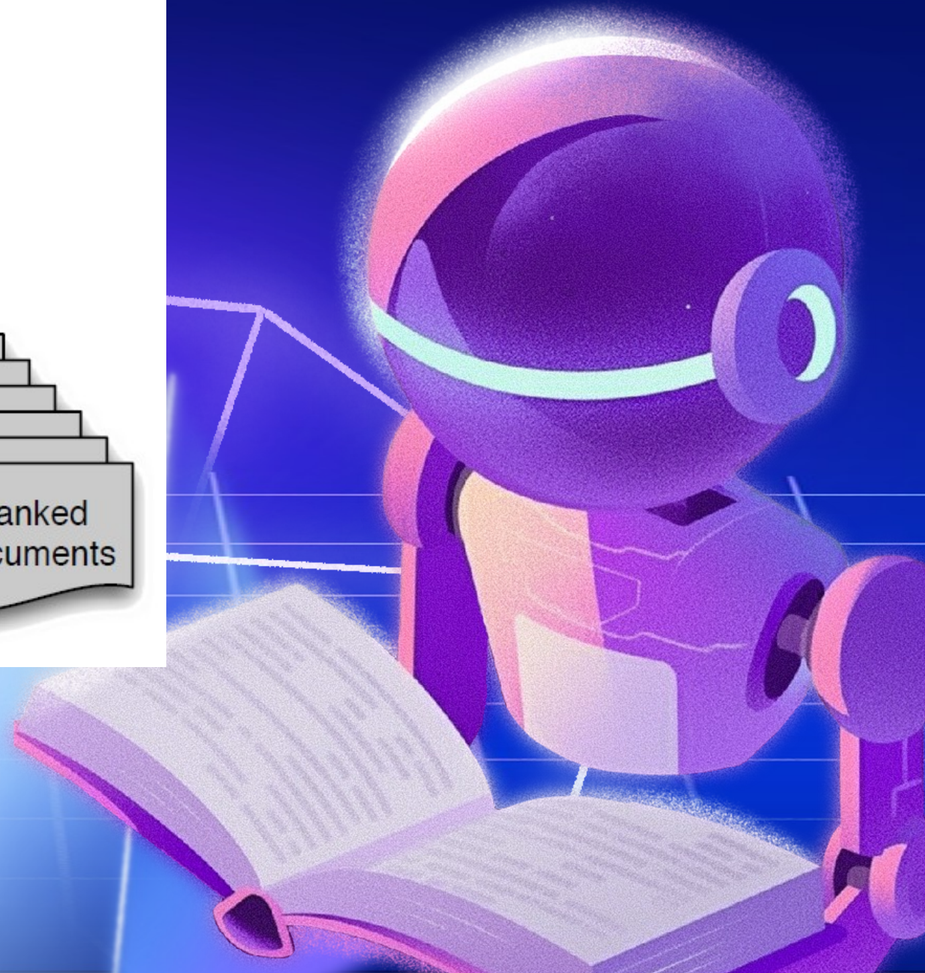
Search Engine Works & History



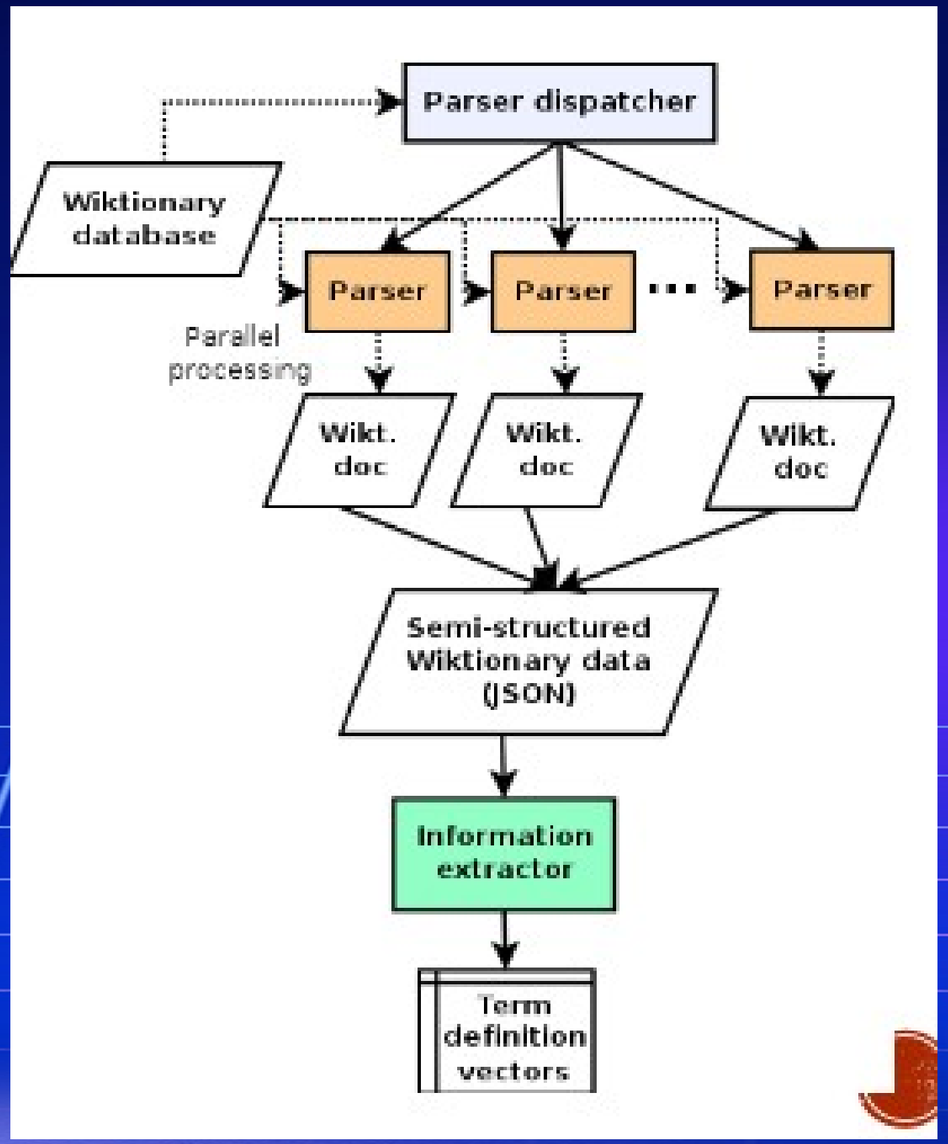


Information Retrieval (IR)



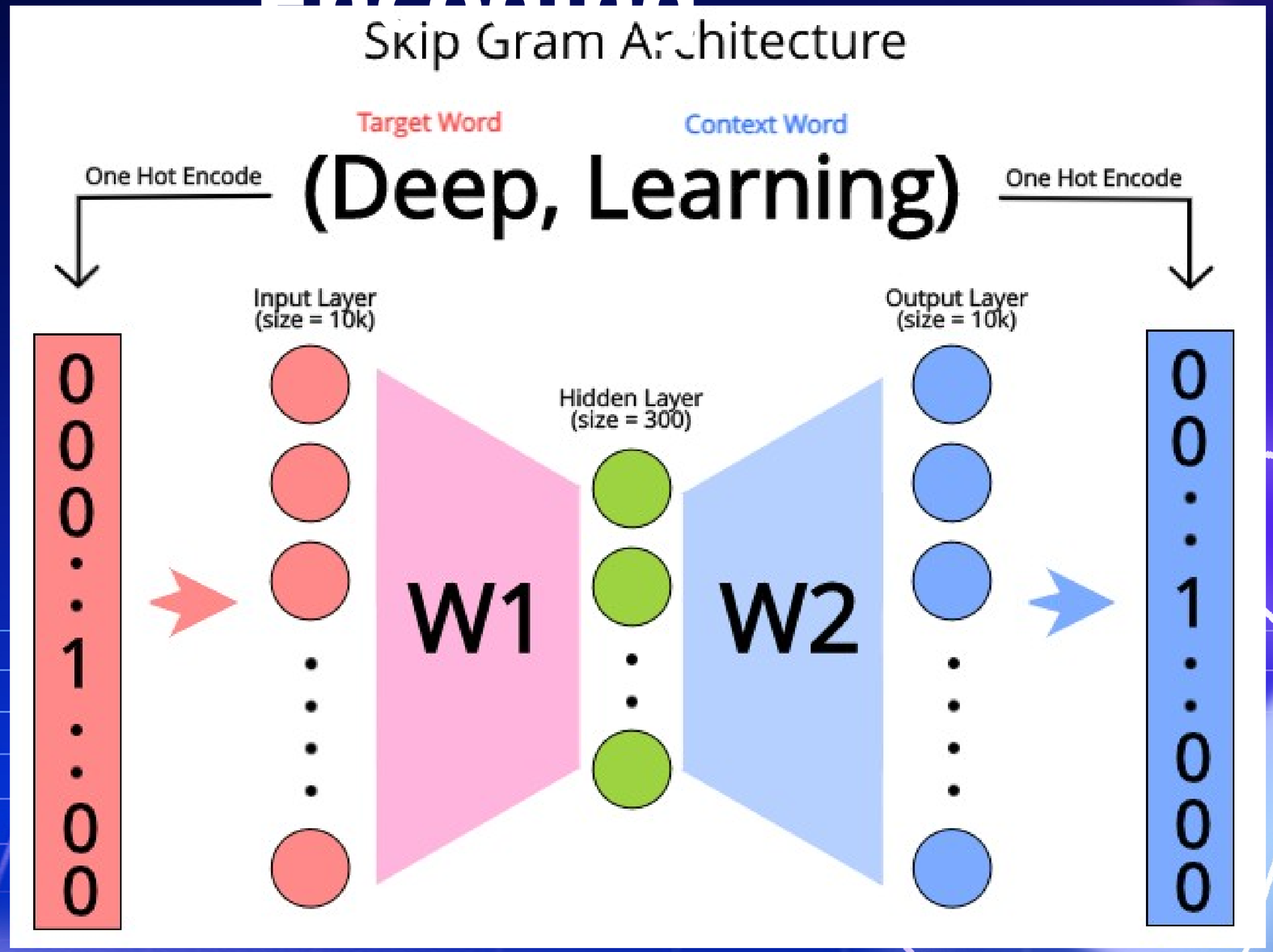


Information Extraction





One Hot Encoding





Word2Vec

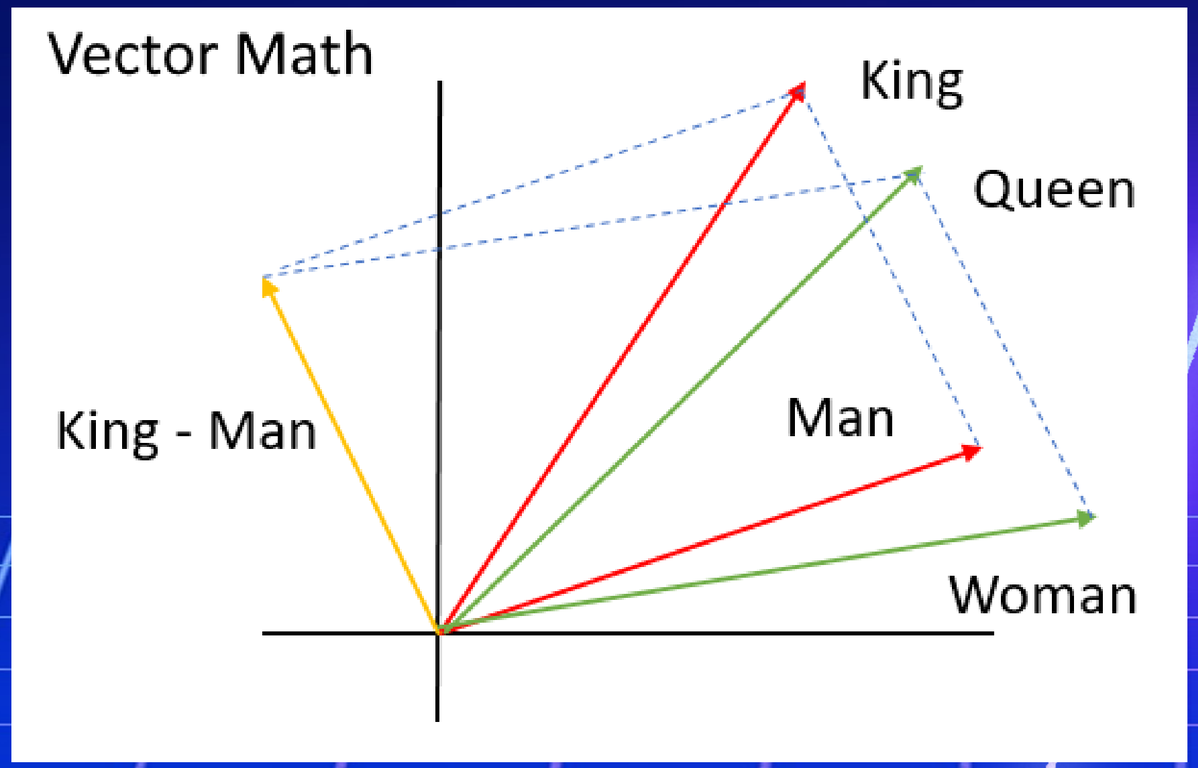
	the	come	Sit	Down
Come	0	1	0	0
Sit	0	0	1	0
Down	0	0	0	1

the	1	come	
Down	3	Sit	

come	Sit	Down
2	4	3

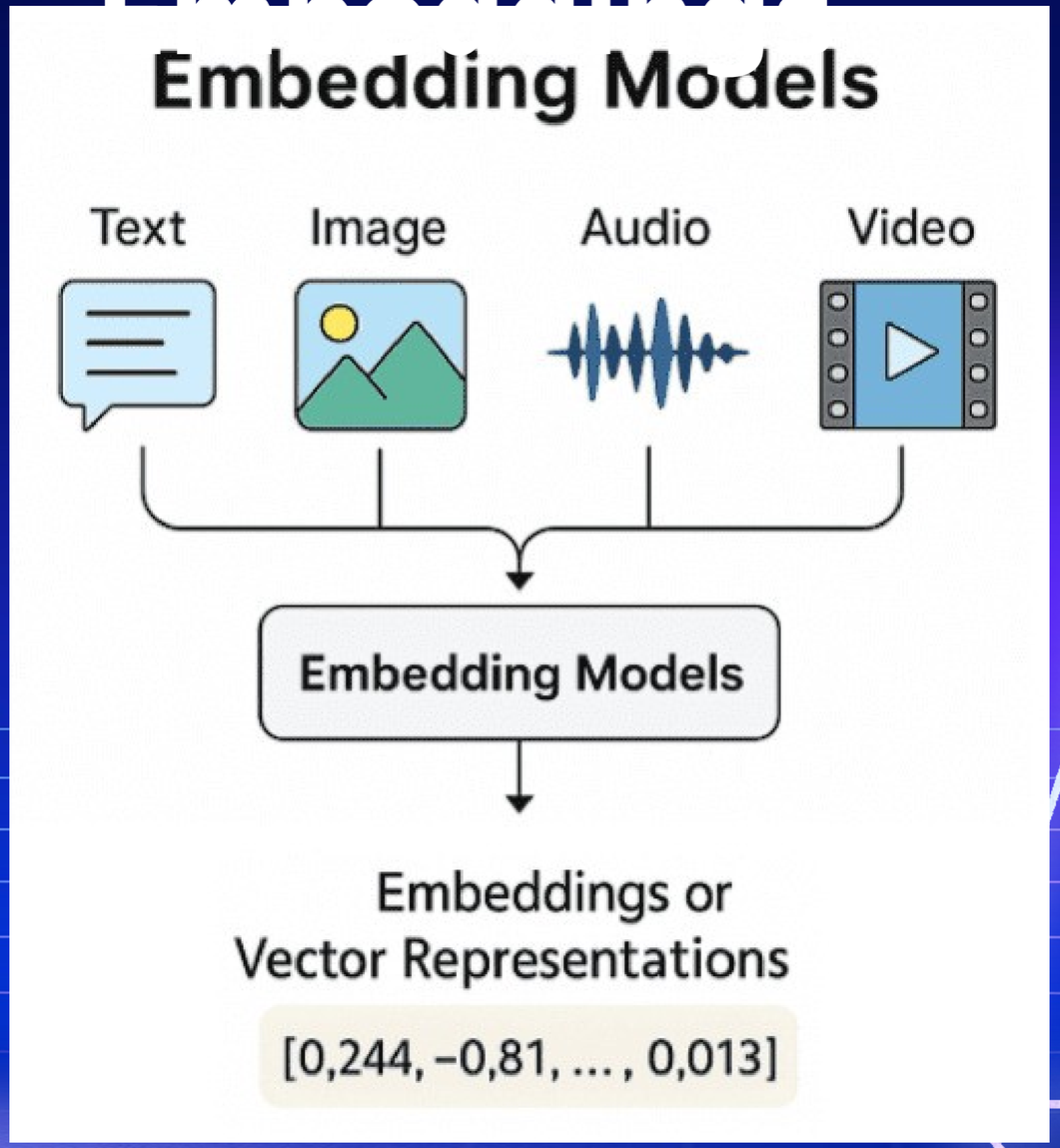


Word Embedding : Word2Vec , Glove



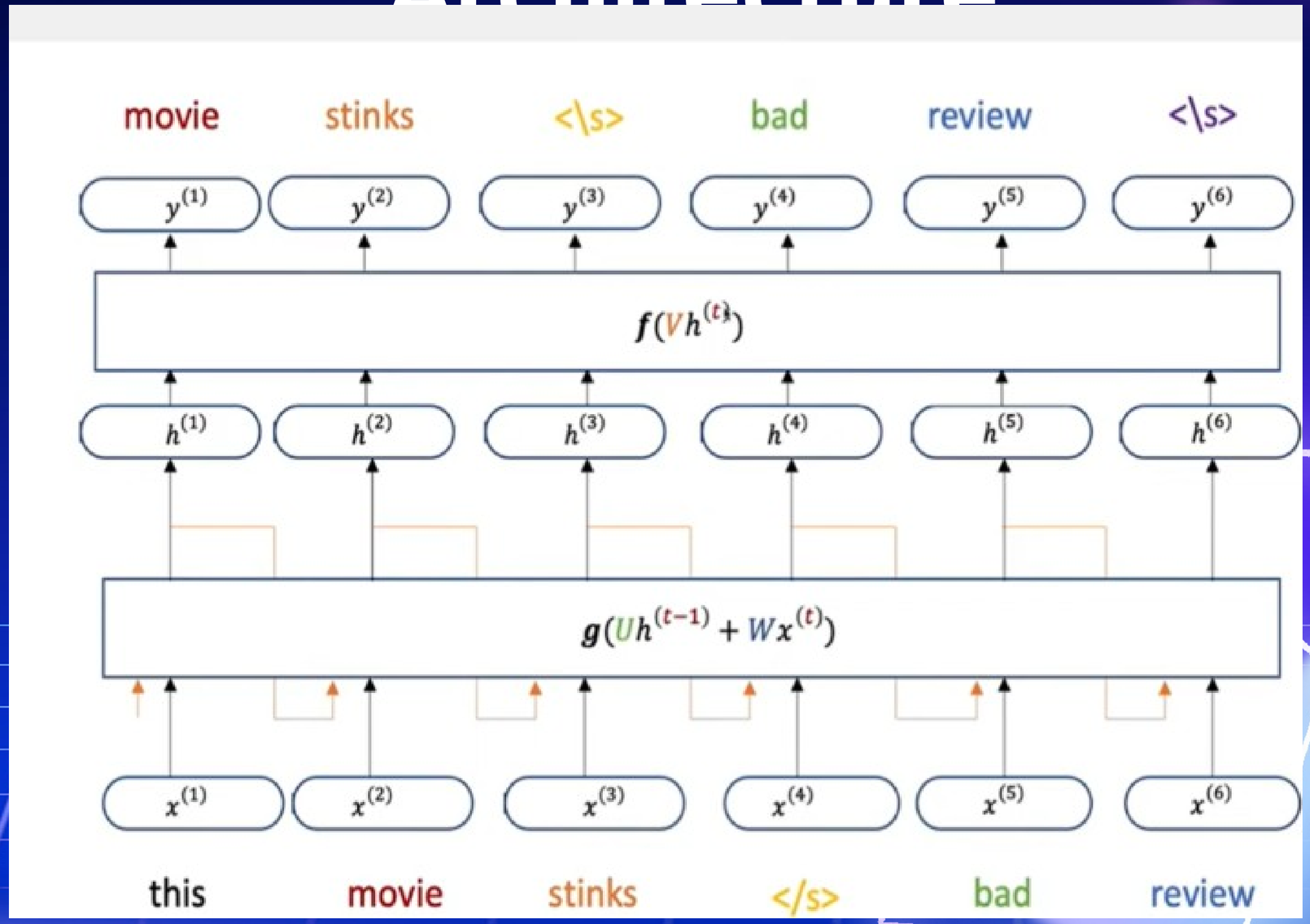


Vector Embeddings



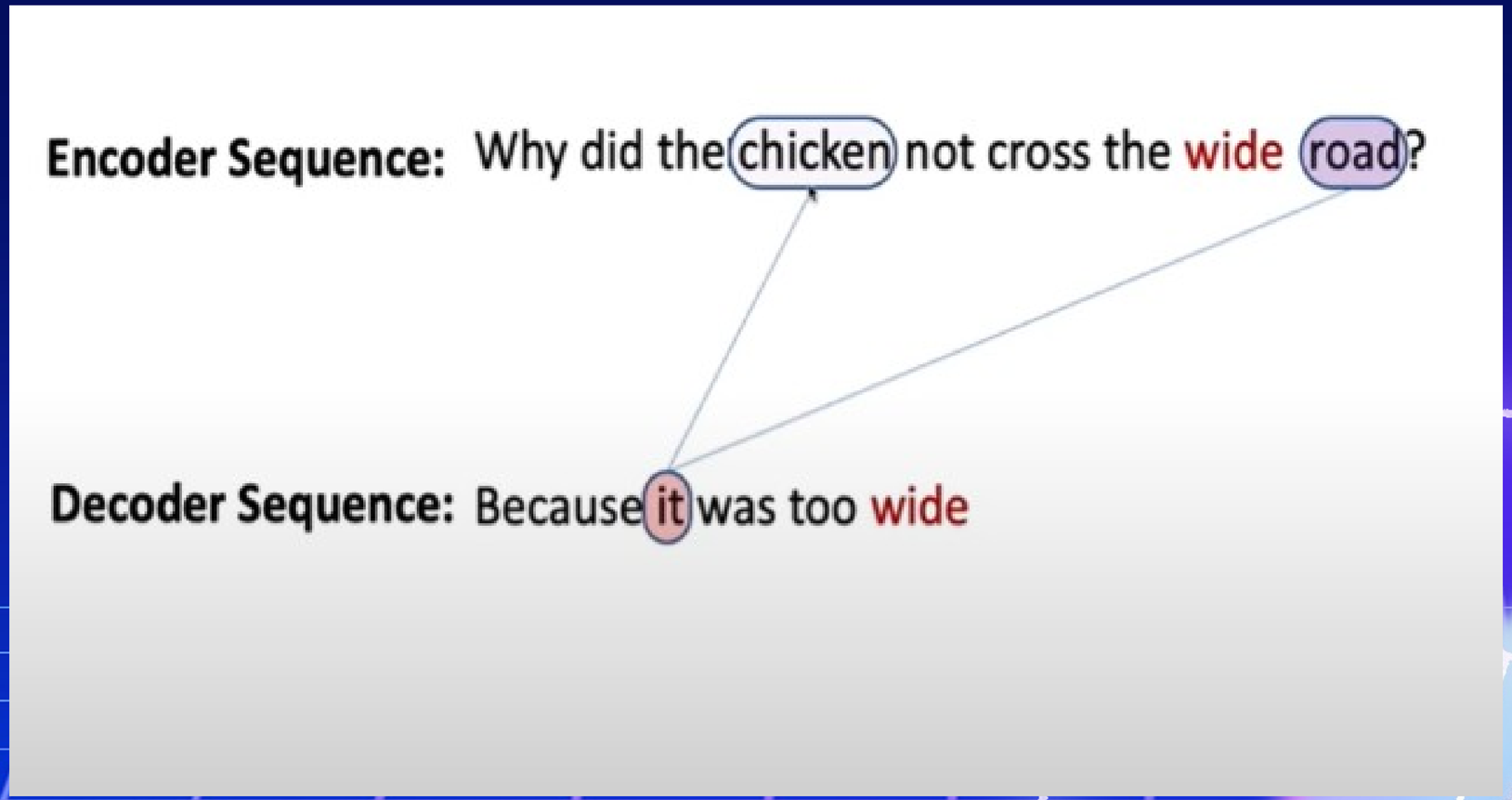


Encoder Decoder Architecture



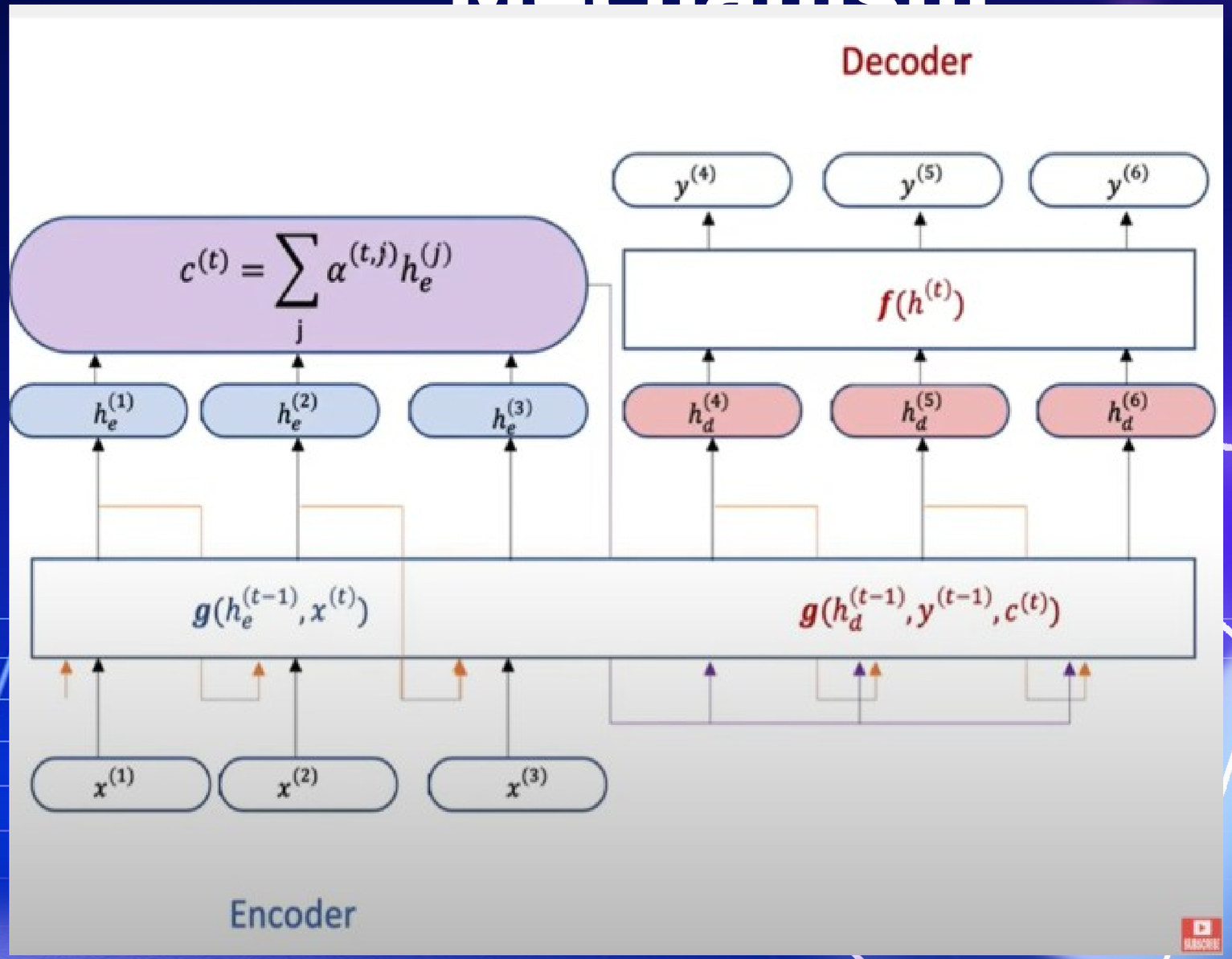


Encoder Decoder Architecture



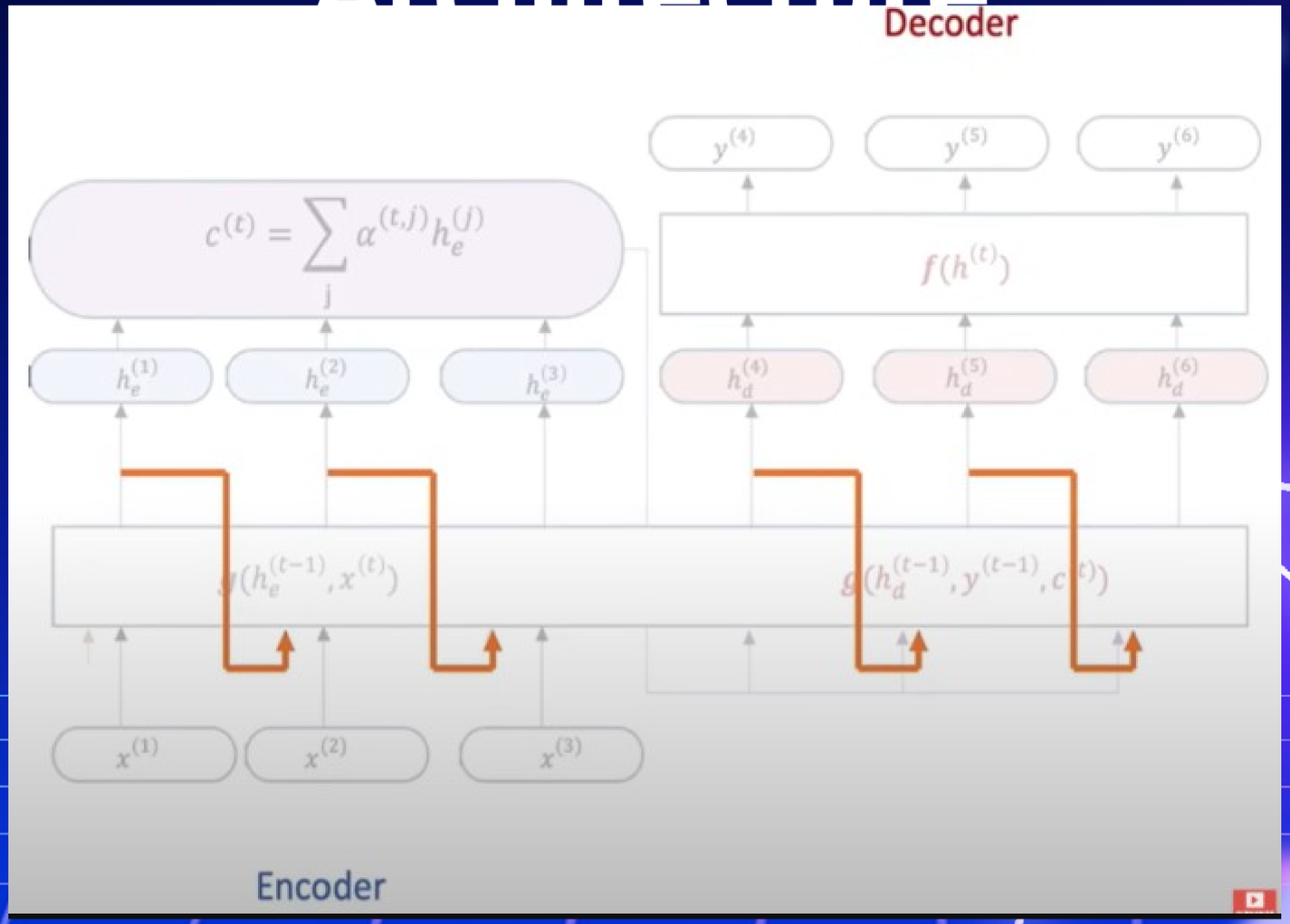


Attention Mechanism



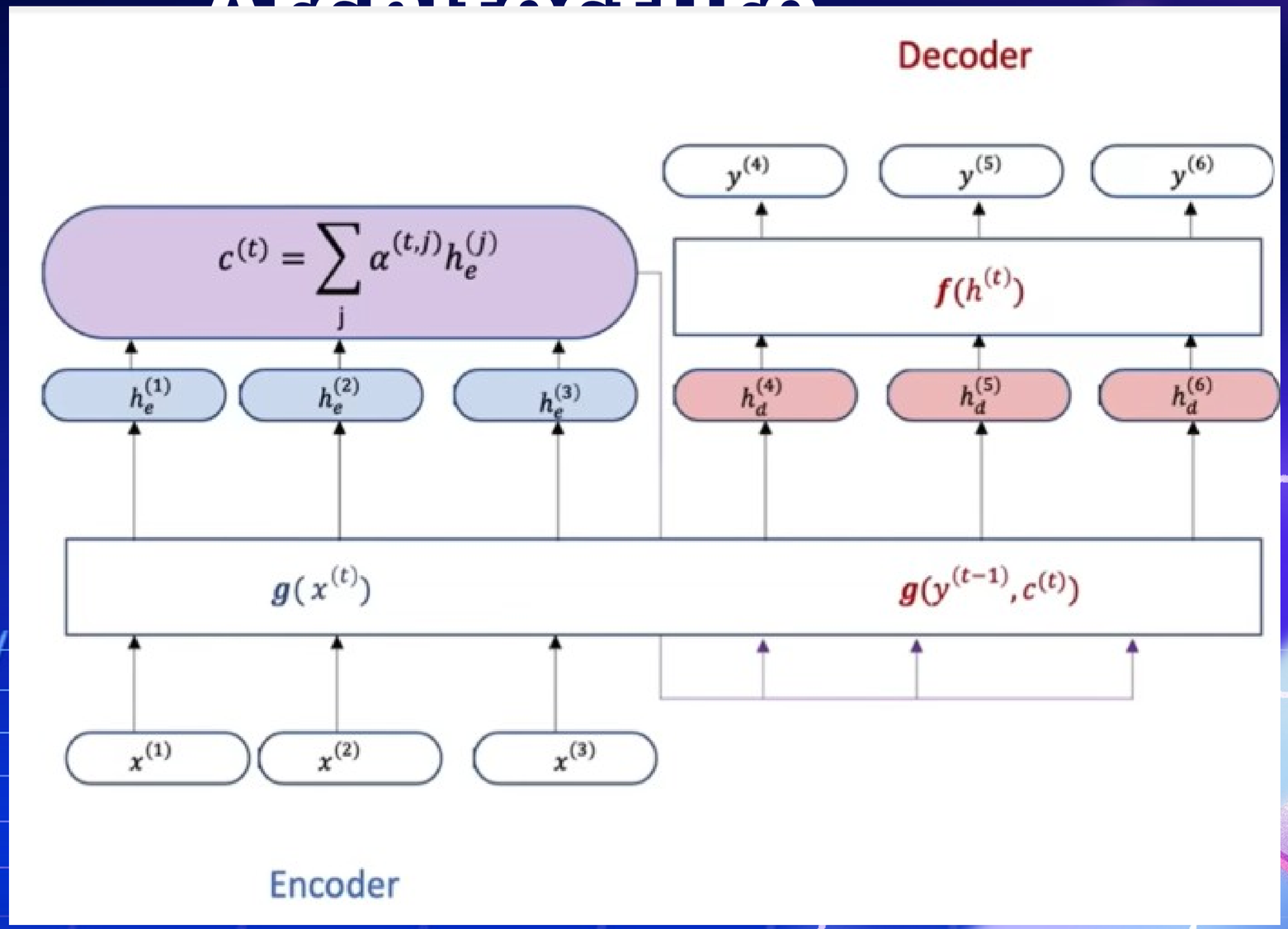


Transformer Architecture





Transformer Architecture



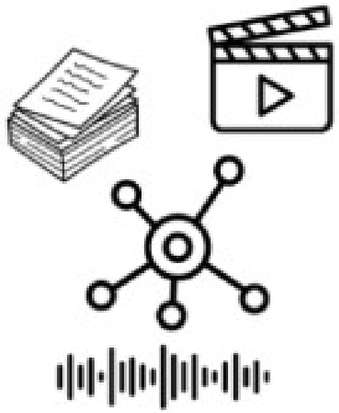


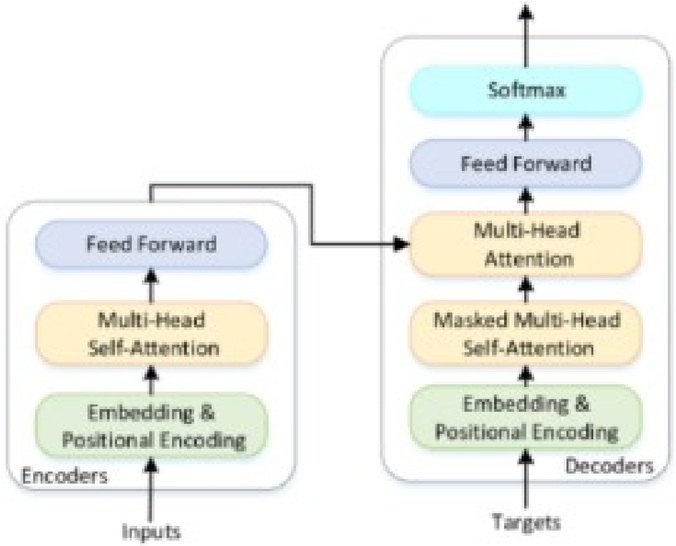
Analytics Vidhya

Evolution of Large Language Models

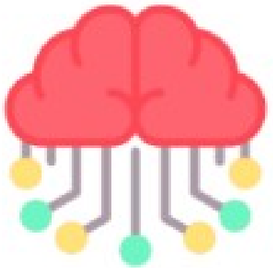


LLM



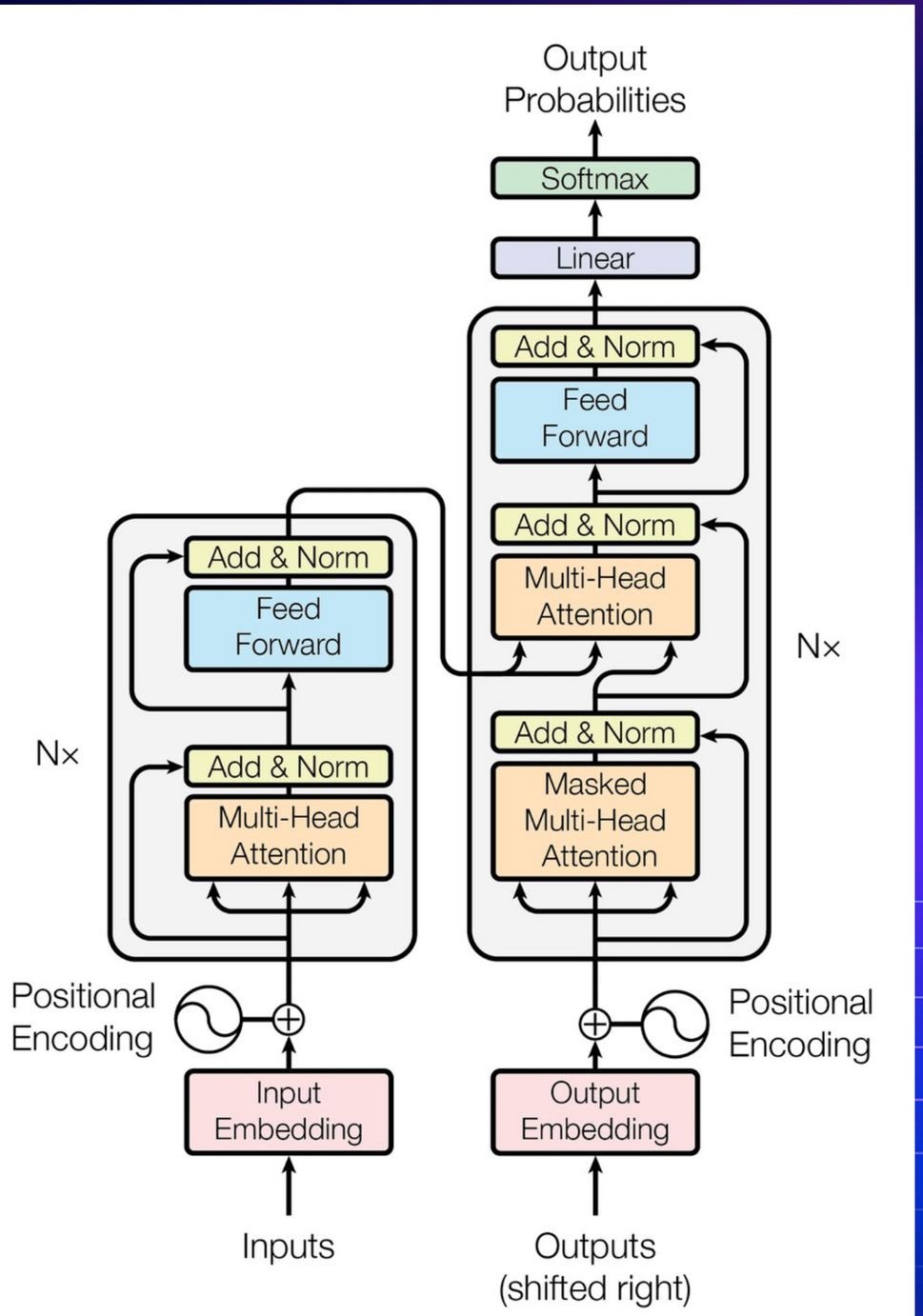






Data + **Architecture** + **Training** = **LLM**

BERT Architecture



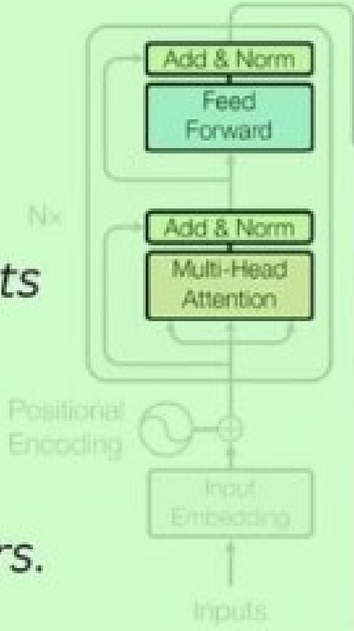


BERT VS GPT

BERT

Google

use transfer learning to ***continue learning*** *from its existing data when adding user-specific tasks and layers.*



GPT

OpenAI

decodes from its massive pre-learned embeddings to present output that matches user prompts. It

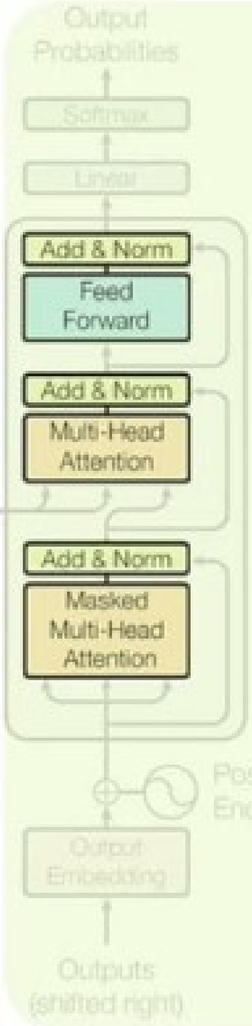
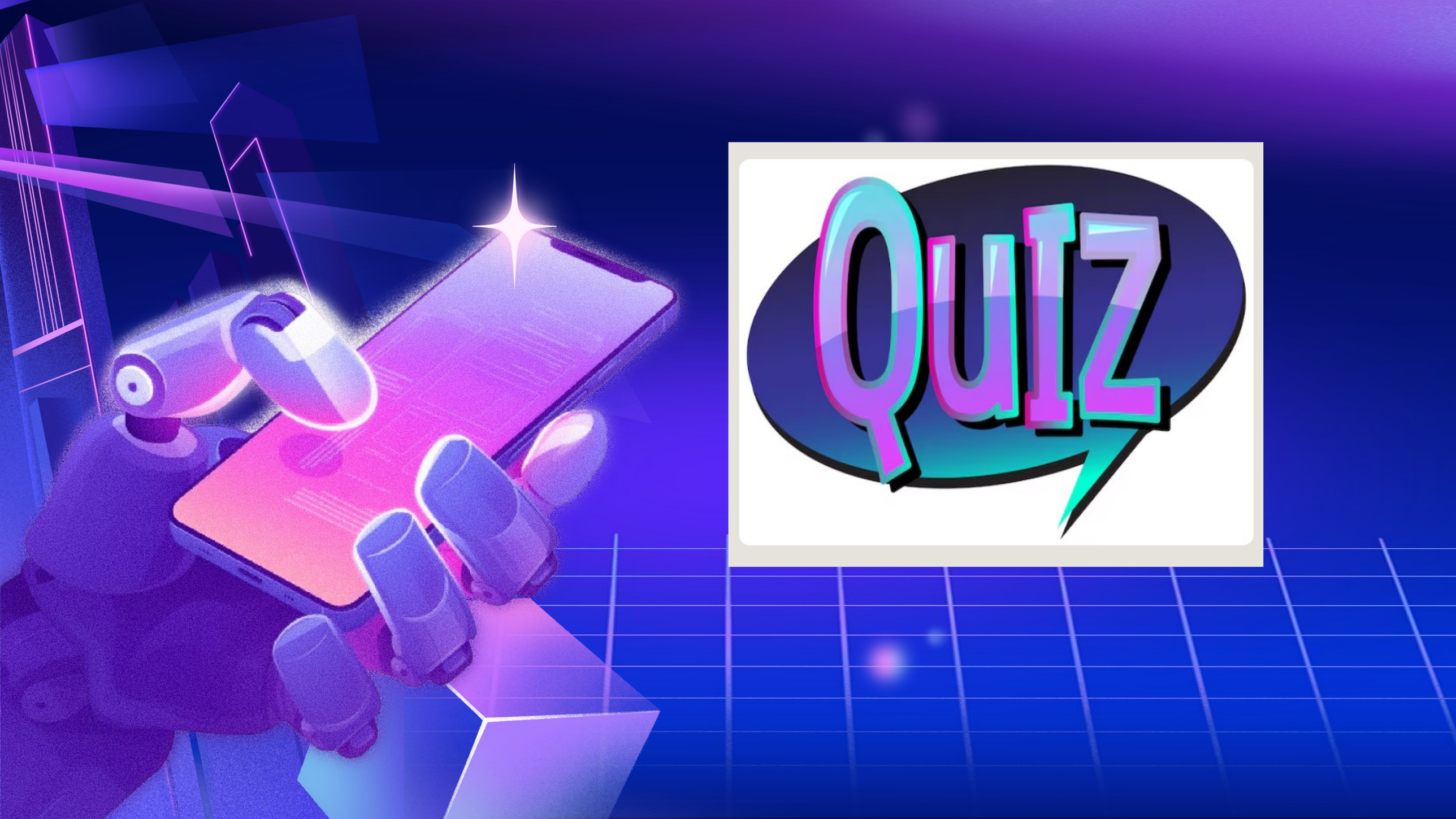


Figure 1: The Transformer - model architecture.


Quiz



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